

Evaluating the Impact of Narrative-Based Augmented Reality on Early STEAM Learning Outcomes in Sri Lankan Primary Education

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Project Proposal Report

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Interactive Media

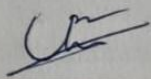
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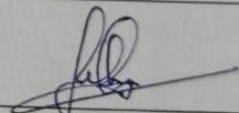
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Sri Lankan Grade 4–5 students consistently underperform in mathematics and science, particularly in topics that require visual, spatial, and process-based understanding. The overall research project develops four independent narrative-based mobile AR storybook applications covering measurement, life science, physical science, and number concepts, each targeting a specific curriculum gap through bilingual Sinhala and English AR content with Art embedded through narrative storybook illustrations, delivered on affordable Android devices. This proposal focuses on Component 3: the AR Physical Science Storybook, which addresses simple machines, energy, and forces topics where student misconceptions are most persistent because the underlying processes are entirely invisible in a conventional classroom. The application uses Unity and Vuforia to render interactive AR scenes triggered by printed storybook pages, allowing students to manipulate 3D representations of simple machines, observe energy transformations, and apply force vectors to AR objects within a locally grounded Sri Lankan narrative. Evaluation will compare student understanding before and after instruction against a control group using conventional textbook methods.

Keywords: Augmented Reality, Primary Education, Physical Science, Simple Machines, Energy, Forces, Sri Lanka, STEAM

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I am grateful to the Department of Information Technology at SLIIT for providing the academic environment and resources that made this research possible. I also wish to thank the Sri Lankan primary school teachers who provided curriculum insights and classroom guidance during early consultations.

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LIST OF ABBREVIATIONS

Abbreviation	Definition
AR	Augmented Reality
SDK	Software Development Kit
STEAM	Science, Technology, Engineering, Art, and Mathematics
SDG	Sustainable Development Goal
FBX	(Filmbox)widely used proprietary 3D file format
API	Application Programming Interface

1. INTRODUCTION

1.1 Background and Context

Physical science topics specifically simple machines, energy, and forces are among the most persistently misunderstood subjects in the Sri Lankan Grade 4–5 Parisaraya (Environmental Science) curriculum. The root cause is conceptual invisibility: the physical processes that govern how simple machines work and how force and energy interact are not directly observable. A lever multiplies force through mechanical advantage, but textbooks can only depict the static outcome of that operation. A force causes an object to accelerate, change direction, or stop, but a static diagram cannot convey direction, magnitude, or the real-time consequence of changing that force. Students aged 9–11 are therefore repeatedly told abstract principles they have no spatial or experiential framework to understand.

The Sri Lankan Ministry of Education Grade 5 Scholarship Examination data consistently identifies physical science as a domain of weak performance. Nationally, student cohorts in Grades 4 and 5 rely almost entirely on printed textbooks and verbal teacher explanation. Schools in provincial and rural areas lack laboratory equipment, models, and multimedia resources. This creates an instructional gap between what the curriculum requires students to understand the mechanical behaviour of simple machines, the relationship between force and energy, and how these concepts manifest in everyday objects and what classroom-based tools can demonstrate. Augmented Reality (AR) has demonstrated substantial learning outcome improvements in comparable contexts globally [1]–[8], yet no AR physical science learning tool exists for the Sri Lankan primary curriculum.

This proposal addresses this gap through the development of a narrative-based mobile AR application specifically designed for the Grade 4–5 Parisaraya physical science curriculum. The application is one of four independent components in a larger research project titled: Evaluating the Impact of Narrative-Based Augmented Reality on Early STEAM Learning Outcomes in Sri Lankan Primary Education. Each component is self-contained and independently evaluable. This component focuses exclusively on light, shadow, and forces using a narrative-driven AR storybook approach. The system aligns with United Nations Sustainable Development Goal 4 (Quality Education) by improving the accessibility of high-quality, contextually relevant STEAM instruction for Sri Lankan primary students, with Art integrated through the narrative storybook illustration design that grounds each AR experience in a culturally familiar visual context.

1.2 Literature Review

The integration of AR in primary science education has gained significant research traction over the past decade. Fokides et al. [1] developed a marker-based AR platform for 5th grade physics in Greek primary schools, demonstrating that AR-based interaction with physical phenomena significantly improved student comprehension scores compared to conventional instruction. The platform overlaid interactive physical science content onto printed physical cards a methodology directly analogous to the storybook-marker approach proposed here, where each printed page triggers a corresponding AR scene.

Cai et al. [2] examined the effects of AR-based physics learning on student self-efficacy and conceptions of learning, finding that students who interacted with AR physics simulations demonstrated not only higher content scores but also significantly improved self-confidence in their ability to learn science. This motivational dimension is particularly relevant for the Sri Lankan primary context, where science anxiety is a documented barrier.

Fidan and Tuncel [3] integrated AR into problem-based learning environments for physics education, reporting statistically significant improvements in learning achievement and attitude. Their study included force-related problem scenarios, where AR force vector visualisations allowed students to see the direction and relative magnitude of forces they were solving for directly informing the interactive force slider design in this proposal.

Mufit et al. [4] conducted a cross-level review of AR in science education, identifying physics as the domain with the most consistent positive effect sizes across studies. Their review emphasised that AR is most effective when it makes normally invisible physical processes wave propagation, particle behaviour, force fields directly visible and manipulable by students.

Zafeiropoulou et al. [5] conducted a pilot study of AR game-based learning in Greek primary school physics, finding that primary-aged students engaged more persistently and developed fewer misconceptions when AR allowed them to directly manipulate a physical variable and immediately observe the resulting change a core interaction design principle applied in this proposal, where students manipulate machine components and force sliders to observe real-time physical outcomes.

Rizki et al. [6] investigated ethnoscience-enhanced physics AR simulation combined with inquiry learning, finding improvements in both student creativity and motivation. The cultural grounding of science content presenting physics phenomena in locally familiar contexts produced stronger learning outcomes than culturally neutral AR content. This finding directly justifies the narrative setting of this proposal (a village in Galle, southern Sri Lanka) as a design choice with evidence-based impact.

Liu et al. [7] conducted a scoping review of VR and AR in K-12 STEM learning with a physics and science focus, identifying key trends including the superiority of mobile over tethered AR platforms for classroom deployment, the effectiveness of combined narrative and simulation modes, and the persistent challenge of scalability in resource-constrained school environments. Their recommendations for low-cost, mobile-first AR design directly shaped the platform choices in this proposal.

Lin et al. [8] conducted a meta-analysis of AR in educational outcomes across 28 studies, reporting a mean effect size of $d = 0.717$ for AR-based instruction compared to conventional methods, with the largest effects observed in science subjects at the primary level. This meta-analytic evidence provides the strongest quantitative justification for the expected learning outcomes of this proposal.

Across this body of literature, three consistent gaps emerge relative to the Sri Lankan primary context: no study has developed AR physical science content anchored in the Sri Lankan curriculum covering simple machines and energy; no AR physics tool has been designed for the Sinhala-speaking primary student population; and no narrative-driven AR approach has been investigated for simple machines, energy, or forces at this educational level in Sri Lanka.

1.3 Research Gap

Despite the global proliferation of AR-based physical science tools, no AR application exists for the Sri Lankan primary Parisaraya curriculum covering simple machines, energy, or forces. The textbooks used by Grade 4–5 students across Sri Lanka contain only two-dimensional static diagrams for these topics, without any interactive or dynamic element. This instructional constraint is particularly damaging for physical science, where student misconceptions are rooted in the inability to visualise process how simple machines multiply or redirect force, why energy transforms from one form to another, how force magnitude and direction affect object motion.

Existing AR physics tools have been developed for secondary and university levels [2], [3], [6], or for non-Sri Lankan primary contexts [1], [5]. None incorporate the specific misconceptions documented in Sri Lankan Grade 4–5 assessment data, and none provide bilingual Sinhala and English content. The narrative-driven approach where a printed story creates the need to observe the physical process before the AR visualises it has not been applied to physical science at the primary level in any South Asian context.

Main Research Question:

Can a narrative-based mobile AR system improve STEAM concept understanding and knowledge retention in Sri Lankan Grade 4–5 students across mathematics and science topics compared to conventional textbook-based instruction?

Sub-Research Questions:

- SRQ1: Does a narrative-driven AR measurement experience improve conceptual understanding and practical application accuracy for measurement topics (length, mass, capacity, time, and data) among Sri Lankan Grade 4–5 students compared to conventional textbook-based instruction?
- SRQ2: Does a narrative AR storybook depicting biological life cycles and ecosystem processes improve scientific process understanding and knowledge retention in Sri Lankan Grade 4–5 students compared to static diagram-based textbook instruction?
- SRQ3: Does an interactive AR physical science storybook improve conceptual understanding of simple machines, energy, and forces and reduce persistent misconceptions in Sri Lankan Grade 4–5 students compared to teacher-led demonstration with static diagrams? (This proposal focuses on SRQ3.)
- SRQ4: Does a narrative AR storybook embedding fraction, decimal, and number operation concepts in a real-world context improve fraction conceptual understanding and number operation accuracy in Sri Lankan Grade 4–5 students compared to conventional symbolic instruction?

This proposal addresses SRQ3 and focuses exclusively on Component 3 the AR Physical Science Storybook.

2. OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate a set of four narrative-based mobile Augmented Reality storybook applications that improve STEAM concept understanding and knowledge retention

among Sri Lankan Grade 4–5 students across mathematics and science topics, where the Art component is expressed through culturally grounded Sri Lankan narrative story arcs and visual aesthetics embedded in the printed storybook design, compared to conventional textbook-based classroom instruction.

2.2 Specific Objectives

1. To develop a narrative-driven AR measurement storybook application that makes length, mass, capacity, time, and data concepts visually concrete for Grade 4–5 mathematics students through interactive 3D measurement instrument simulations. (Component 1)
2. To develop a narrative-driven AR life science storybook application that incorporates biological processes plant life cycles, food chains, and ecosystems using locally grounded Sri Lankan species to improve scientific understanding in Grade 4–5 Parisaraya students. (Component 2)
3. To develop a narrative-driven AR physical science storybook application that makes simple machine mechanics, energy transformations, and force dynamics spatially visible and interactively manipulable for Grade 4–5 Parisaraya students, targeting the misconceptions that arise from static, diagram-only instruction. (Component 3: this proposal)
4. To develop a narrative-driven AR number concepts storybook application that embeds fractions, decimals, place value, and number operations within a real-world bakery context to improve number concept understanding and operational accuracy in Grade 4–5 mathematics students. (Component 4)

3. METHODOLOGY

3.1 Research Approach

This research follows a build-and-test approach: the system is first designed and developed based on identified learning needs, and then tested with real students to measure its effectiveness. The evaluation involves two groups of Grade 4–5 students one group uses the AR storybook application, and the other group receives the same topic through conventional textbook instruction. Both groups are tested before and after the session using the same questions, and the results are compared to determine whether the AR approach led to better understanding.

3.2 Technical Methods and Tools

- Unity 3D: The primary development environment for building the AR scenes, 3D models, physics simulations (shadow casting, force arrows), and the Narrative Manager script that controls which scene appears for each storybook page.
- Vuforia Engine SDK: Used to set up each printed storybook page illustration as a scannable marker. When the student points the camera at a page, Vuforia recognises it and Unity loads the correct AR scene.
- Blender: Used to create 3D models for the story environment the village setting, torch, bicycle, ball, and other scene props.
- Adobe Illustrator: Used to design the storybook page illustrations, making sure each page works both as a readable story image and as a high-quality scannable marker.
- Firebase: Used to record anonymised usage data such as which pages students scanned, how they interacted, and whether their answers were correct.

3.3 Data Collection and Ethical Considerations

Participants will be Grade 4–5 students from selected government primary schools in Sri Lanka. Written parental consent and school administration approval will be obtained before any data collection begins. Student identities will be anonymised in all records. Tests will be conducted in Sinhala so that language does not affect performance. All data will be stored securely and used only for academic research, in line with SLIIT Research Ethics Committee guidelines.

3.4 Validation and Performance Metrics

- The main measure of success is whether students in the AR group score noticeably better in the post-test compared to the pre-test, and compared to the control group who used conventional instruction.
- Misconception tracking: Specific wrong-answer patterns related to simple machines, energy, and forces will be monitored before and after the session to check whether the AR experience helped correct common misunderstandings.
- Engagement: Firebase data will show how long students spent on each page and whether they completed the full storybook session.
- Usability: A simple, age-appropriate usability questionnaire will be given to students after using the app to check if it was easy and enjoyable to use.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

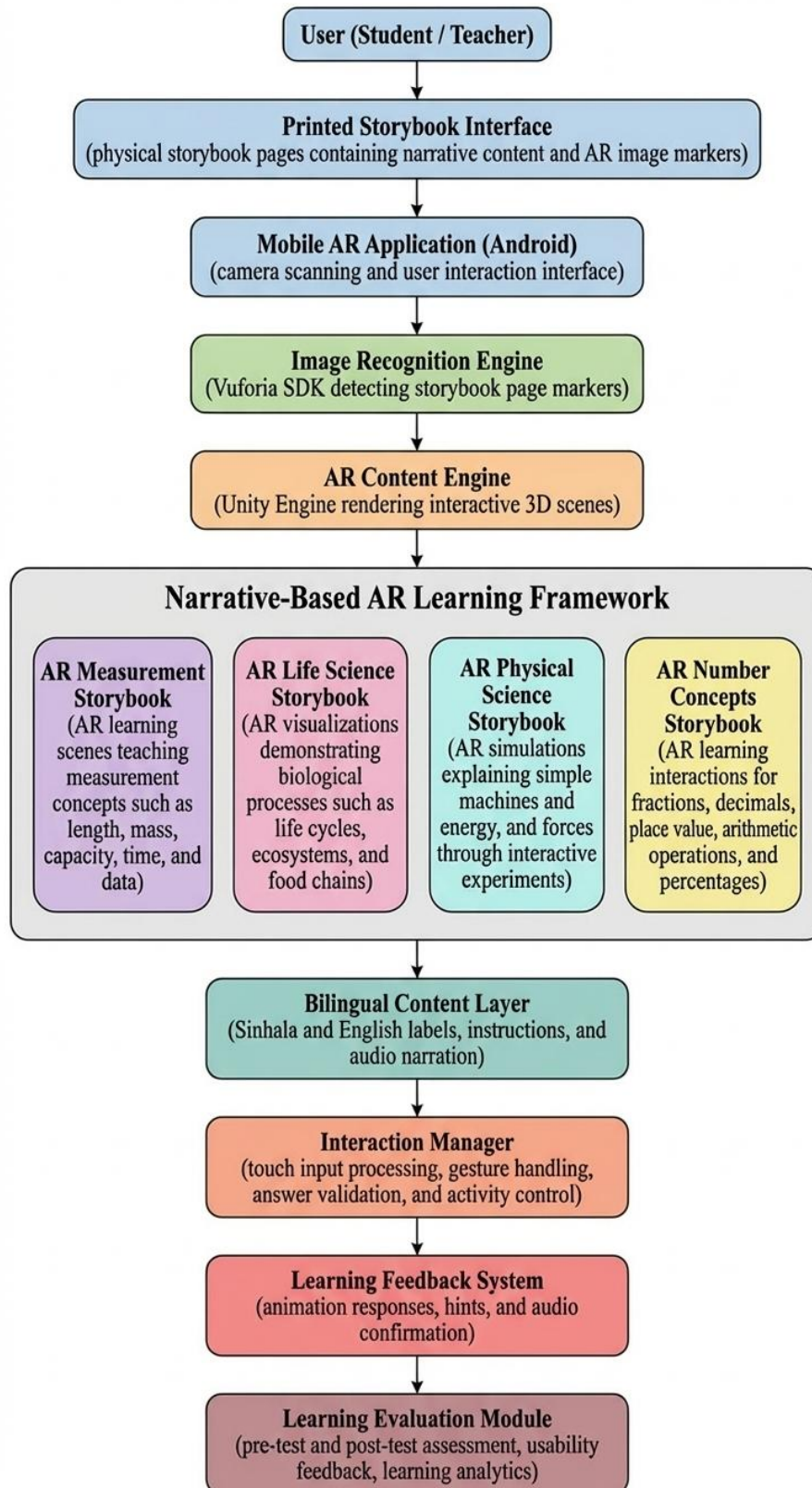
4.1 Overview

The system architecture consists of three primary layers: the Physical Layer (the printed storybook), the AR Application Layer (Unity + Vuforia mobile application), and the Backend Analytics Layer (Firebase). These layers interact as follows: a student scans a storybook page using their Android device camera; Vuforia detects the Image Target marker on that page; Unity's Narrative Manager script loads the corresponding AR scene for that page's narrative stage; the AR scene renders on a flat surface in front of the student and waits for interaction input; interaction results (correctness, time taken) are logged to Firebase anonymously.

The system is offline-capable for all core AR functions. Firebase synchronisation occurs when internet connectivity is available, ensuring the application is usable in schools with unreliable internet. The application targets Android devices to cover the affordable smartphone range common in Sri Lanka.

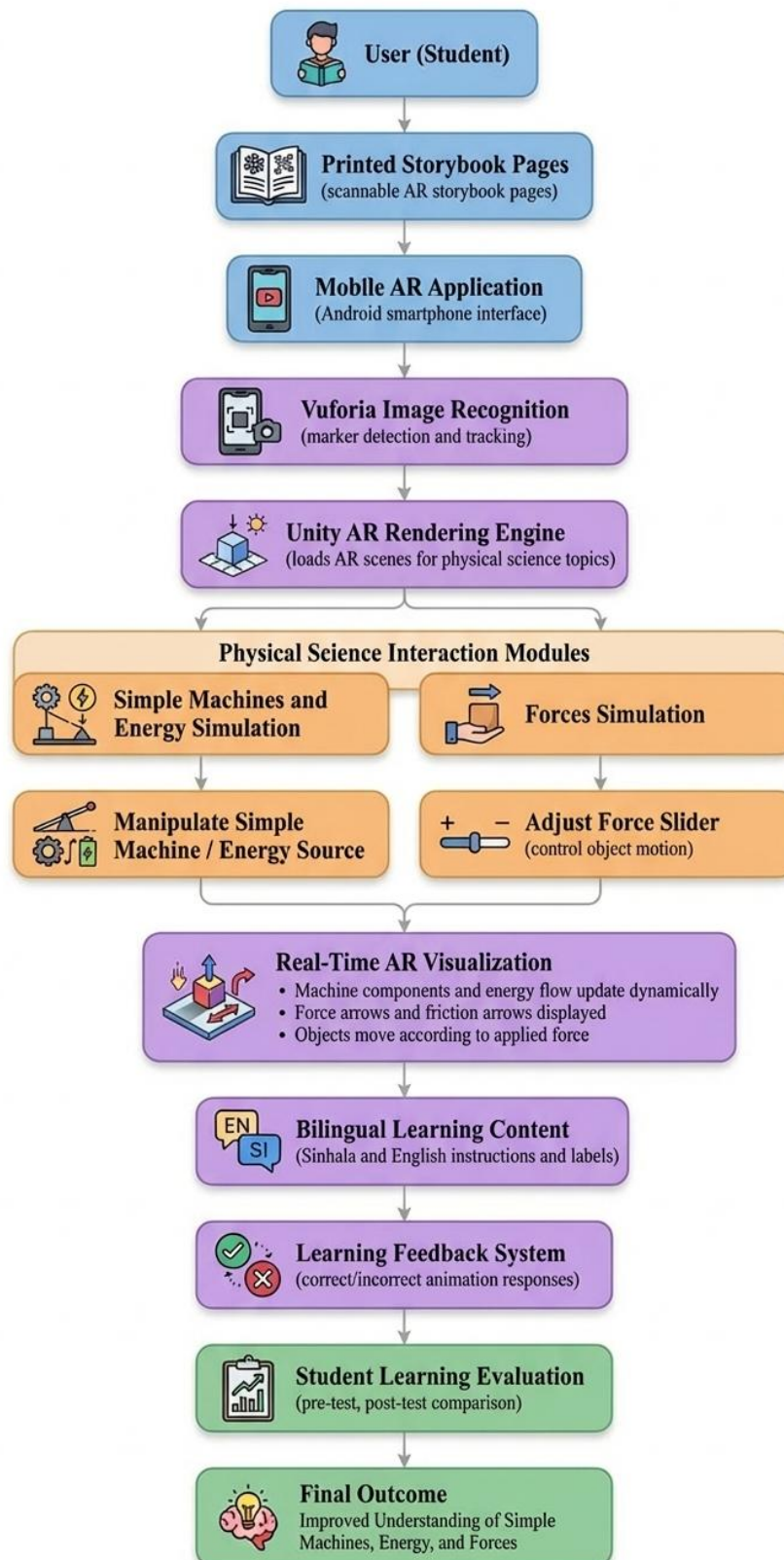
4.2 System Architecture Diagram

Figure 1: High-Level System Architecture Diagram



4.3 AR Physical Science Storybook Architecture Diagram

Figure 2: AR Physical Science Storybook System Diagram



4.4 Component Interactions and Data Flow

- **Physical Layer:** Six storybook pages, each with a distinct high-contrast Vuforia ImageTarget illustration. Each page corresponds to one narrative stage and one AR scene.
- **Vuforia Engine:** Continuously processes the camera feed. On marker detection above a threshold confidence score, it returns the marker ID and pose (position and orientation) to Unity.
- **Unity Narrative Manager:** A C# script that maps marker IDs to scene configurations. On receiving a marker ID, it instantiates the correct AR scene prefab, positions it relative to the marker pose, and activates the interaction components for that scene.
- **AR Scene Components (per page):** Simple Machines & Energy scenes contain interactive machine GameObjects and an EnergyTransfer script. Forces scenes contain a 3D Rigidbody object, a ForceSlider UI element, and FrictionArrow and ForceArrow GameObjects.
- **Interaction Handler:** Detects student touch input (drag gesture for machine components, slider gesture for force application) and updates the corresponding physics simulation in real time.
- **Feedback System:** After each interaction, an AnswerChecker script compares student input against the correct value (machine output, force outcome) and triggers either a positive AR visual feedback (green glow, correct label) or a prompt-to-retry.
- **Firebase Analytics Module:** Sends anonymised session events (marker ID, interaction type, result, duration) to Firebase Realtime Database when network is available.

4.5 Scalability and Security Considerations

The application is designed for offline-first operation, ensuring full functionality without internet connectivity. Firebase synchronisation is non-blocking; analytics uploads are queued locally and sent when connectivity is restored. All student data transmitted to Firebase is anonymised at source no personally identifiable information is collected. The application can be extended to additional AR scenes without architectural changes by adding new ImageTarget images and scene prefab configurations to the Narrative Manager map.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

- Primary Users: Sri Lankan Grade 4–5 students as the direct learners interacting with the AR storybook.
- Secondary Users: Primary school teachers who guide students through the storybook sessions and administer pre/post-tests; and parents of Grade 4–5 students who may supervise use at home and are involved in the consent process.
- Institutional Stakeholders: School administrators responsible for device provisioning and parental consent management.
- Research Stakeholders: SLIIT academic supervisors and the research team evaluating outcomes.

5.2 Functional Requirements

- The system shall detect and track each of the six storybook page ImageTarget markers using the Android device camera.
- The system shall render the correct AR scene on a flat surface in front of the student within an ideal response time after successful marker detection.
- In Simple Machines & Energy mode, the system shall allow the student to interact with 3D machine components and observe the resulting energy transfer and
- In Forces mode, the system shall allow the student to use an on-screen slider to increase or decrease the applied force magnitude and display force arrows and object motion accordingly.
- The system shall display all text labels in both Sinhala and English, switchable by the user.
- The system shall provide immediate visual feedback (correct/incorrect visual feedback) after each student interaction.

- The system shall function without internet connectivity for all core AR and interaction features.
- The system shall log anonymised session data to Firebase when internet connectivity is available.

5.3 Non-Functional Requirements

- Performance: AR scene rendering shall maintain a smooth and consistent frame rate suitable for interactive use on mid-range Android devices commonly available in Sri Lanka.
- Usability: The application shall be straightforward and intuitive enough for Grade 4–5 students to use without prior training, verified through age-appropriate usability testing.
- Accessibility: All interactive UI elements shall be large enough for comfortable touch use by primary school-aged children; the interface shall default to Sinhala as the primary language.
- Security: No personally identifiable student data shall be collected or transmitted. Firebase access shall be secured with appropriate authentication.
- Reliability: The application shall remain stable during a full storybook session. Marker detection shall function reliably under standard indoor classroom lighting conditions.
- Portability: The application shall run on Android devices without requiring device-specific configuration.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

Primary market: Sri Lankan government and private primary schools seeking supplementary STEAM learning tools for Grades 4–5. Secondary market: Tuition classes and private educational institutes that prepare students for the Grade 5 Scholarship Examination, where physical science is a tested subject. Tertiary market: Parents of Grade 4–5 students looking for interactive home learning tools for science preparation. Extended market: EdTech distributors serving South and Southeast Asian primary education sectors with bilingual AR content requirements.

6.2 Value Proposition

The application provides the only AR physical science learning tool designed specifically for the Sri Lankan Grade 4–5 Parisaraya curriculum, available in both Sinhala and English, deployable on affordable Android smartphones without internet dependency, and validated through empirical student outcome data.

6.3 Revenue Model

- School and Institute Licensing: Annual license fee for schools, tuition centres, and educational institutes covering unlimited student use within one institution.
- Storybook Sales: Printed storybooks sold as physical consumables per student, creating a recurring revenue stream independent of the software license.
- Content Expansion Packs: Additional AR scene packs for other Parisaraya topics sold as add-on licenses to existing institutional customers.

6.4 Cost Estimation

The initial development cost covers software licenses, hardware for testing, storybook design and printing, and developer labour across the research phase. After the research phase, ongoing costs are expected to be low primarily covering cloud hosting, periodic content updates, and storybook reprinting for new student cohorts. The printed storybook is the main recurring physical cost, as each student requires their own copy to use as an AR marker. Digital distribution of the application itself through the Google Play Store has a one-time registration fee, after which updates can be pushed at no additional distribution cost.

6.5 Pricing Strategy

- **Institutional License (Schools and Tuition Institutes)** A flat annual license fee per institution, covering unlimited student use of the application within that institution. This makes the cost predictable for school administrators and removes any per-student barrier to access.
- **Storybook Sold separately as a physical printed book per student.** Keeping the storybook affordable is important since it is a required physical component of the system pricing will be set to be comparable to existing school workbooks available in the Sri Lankan market.
- **Freemium Entry Option:** A free version of the application covering one storybook chapter may be offered to schools and institutes to demonstrate the product before committing to a full license, lowering the adoption barrier.
- **Bundle Pricing** Schools that adopt all four components of the broader AR STEAM system (Measurement, Life Science, Physical Science, and Number Concepts) would be offered a discounted bundle rate compared to purchasing each component license individually.

6.6 Competitive Advantage

No direct competitor exists for AR physical science instruction in the Sri Lankan primary market. The combination of curriculum alignment, bilingual content, storybook narrative design, and offline capability creates a defensible market position. The empirical validation component provides evidence-based differentiation from unvalidated edtech tools.

6.7 Intellectual Property Considerations

The storybook narrative, character designs, and AR scene assets are original works created for this project. Copyright will be registered with the National Intellectual Property Office of Sri Lanka. The Unity + Vuforia technical stack is used under standard commercial licenses. A patent application for the narrative-AR page-trigger interaction design may be considered upon completion of the research phase.

7. BUDGET AND JUSTIFICATION

The following table presents the estimated project budget with justification for each line item.

Table 1: Budget and Justification

Item	Description	Estimated Cost (LKR)
Software Licenses	Unity(student), Vuforia Engine SDK	0
Software Licenses	3D modeling Blender	0
Cloud Services	Firebase (backend/analytics), Google Play deployment	15,000
Human Effort	Developer labour cost for design, development, and testing	270,000
Printing & Materials	Storybook printing, testing stationery	10,000
Miscellaneous	Transport, internet, contingency	8,000
Total		303,000

8. WORK BREAKDOWN STRUCTURE (WBS)

1. Literature Review and Domain Research

Review existing research on AR in physical science education, identify key papers on simple machines, energy, and forces instruction, and study the Sri Lankan Grade 4–5 Parisaraya curriculum requirements to ensure all content is curriculum-aligned.

2. System Architecture Design

Define the overall structure of the application including the component layers (storybook, AR application, analytics backend), the data flow between them, and the interaction model for each AR scene.

3. Storybook Narrative and Scene Design

Write the bilingual (Sinhala and English) narrative for all six storybook pages following the Dasun's Machine Discovery story arc. Design the page layouts so that each illustration serves as both a readable story image and a scannable Vuforia marker.

4. 3D Asset and Environment Creation

3D Asset and Environment Creation: Create all 3D models and assets required for the AR scenes using a combination of approaches. Primary models are built directly in Blender, while additional assets are sourced through Meshy AI generation and existing 3D model libraries. All externally sourced and AI-generated assets are modified and adapted to match the visual style, scale, and interaction requirements of the storybook narrative before integration into the AR scenes.

5. Unity and Vuforia AR Environment Setup

Set up the Unity project, configure the Vuforia SDK, import all 3D assets, and establish the Narrative Manager script that maps each storybook page marker to its corresponding AR scene.

6. AR Scene Development: Simple Machines, Forces & Energy Module

Build the three AR scenes corresponding to the simple machines and energy pages of the storybook. Implement interactive 3D machine components with real-time energy transfer visualisations and develop the drag-and-tap interaction that allows students to operate machine parts and observe the resulting mechanical output and energy changes.

7. AR Scene Development: Forces Module

Build the three AR scenes corresponding to the forces pages of the storybook. Implement the force slider interaction, force arrow and friction arrow GameObjects, and the physics-driven object motion that responds to student input.

8. Bilingual Interface Integration

Integrate all Sinhala and English text labels, instructions, and feedback messages throughout the application. Implement the language toggle feature that allows switching between Sinhala and English during a session.

9. Feedback and Interaction Logic

Develop the AnswerChecker logic that evaluates student responses for each page and triggers the appropriate visual feedback (correct or retry prompt).

10. Firebase Analytics Integration

Set up the Firebase project, implement the anonymised session logging module, and test data synchronisation to ensure usage data is captured and uploaded correctly when internet connectivity is available.

11. Evaluation Instrument Design

Design the pre-test and post-test question sets in Sinhala targeting the specific simple machines, energy, and forces misconceptions identified in the literature. Design the age-appropriate usability questionnaire for student feedback.

12. Pilot Testing with Grade 4–5 Students

Conduct the evaluation study in selected primary schools, administering pre-tests, running AR storybook sessions with the experimental group and conventional instruction with the control group, and collecting post-test results.

13. Data Analysis and Results Documentation

Analyse the pre-test and post-test results to determine whether the AR group showed improved understanding compared to the control group. Document findings, limitations, and recommendations.

14. Final Report and Submission

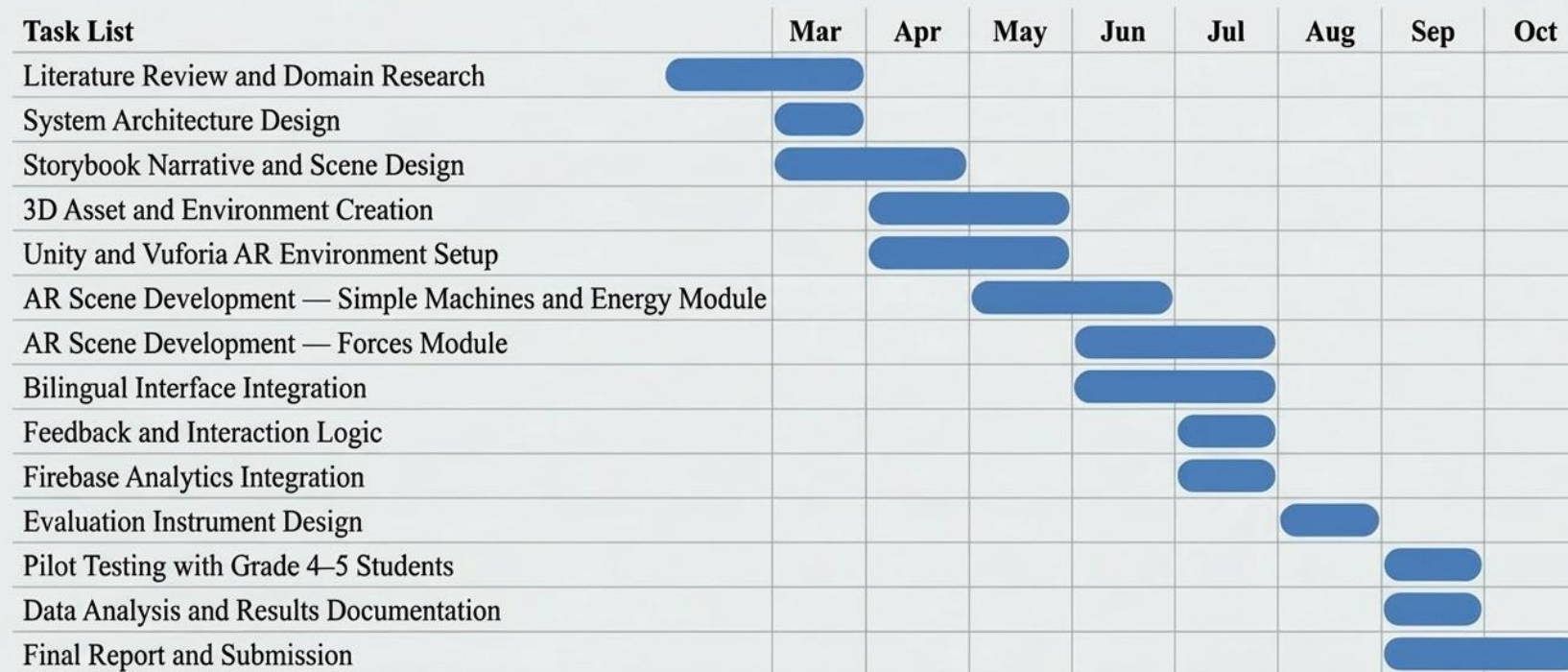
Compile all research documentation, system documentation, evaluation results, and supporting materials into the final project report and prepare the submission package.

9. GANTT CHART

Figure 3: Gantt Chart

Semester Timeline – Gantt Chart

This Gantt chart illustrates the development timeline for the AR Physical Science Storybook system focusing on simple machines and energy concepts for Sri Lankan Grade 4–5 Parisaraya education. The project includes research, AR development, testing, classroom evaluation, and final dissertation preparation between February and October.



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